



Mahaveer Regar

B.Tech in Chemical Engineering

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EDUCATION

B.Tech in Chemical Engineering	IIT Hyderabad	2025	7.42
XII (RBSE)	Gyandev Public Sr. Sec. School, Sikar	2020	98.20%
X (RBSE)	Gyandev Public Sr. Sec. School, Sikar	2018	93.83%

SKILLS

- **Machine Learning:** Supervised ML, Unsupervised ML, Ensemble Learning, Recommender Systems
- **Deep Learning:** ANN, CNN, RNN, LSTM, NLP, Attention Mechanism, Transformers, GPT-2 Architecture.
- **Data Structures:** Arrays, Linked Lists, Stacks, Queues, Binary Trees, Dictionary, Graphs.
- **Algorithms:** Sorting, Searching, Graph Algorithms, Dynamic Programming, Complexity Analysis
- **GenAI:** Retrieval-Augmented Generation (RAG), LLMs, FAISS, ChromaDB, LangChain
- **Libraries:** Scikit Learn, Pandas, Numpy, Matplotlib, Keras, Tensorflow, PyTorch, NLTK.

RELEVANT COURSES

- Introduction to Programming, Data Science and Analysis, Machine Learning for Process Systems.
- Numerical Methods, Advance Numerical Methods, Calculus, Linear Algebra, Applied Mathematics.

PROJECTS

Movie Recommender System

2024

- A web application that offers personalized movie recommendations by analyzing user preferences and historical movie data.
- It uses vectorization techniques like Bag of Words to convert movie keywords into vectors. Using Cosine Similarity it recommend movies.

Tech Stack: Pycharm(IDE), Python(language), NLTK, scikit-learn, Numpy, Pandas, Streamlit)

RAG-based Document Chatbot using Groq LLaMA3

2025

- Developed a modular Retrieval-Augmented Generation (RAG) application enabling users to upload PDFs and interact with an AI chatbot that answers queries based on document content.
- Designed a microservice architecture with a decoupled FastAPI backend and Streamlit frontend for better scalability and maintainability documents Integrated vector database for efficient document retrieval.

Tech Stack: FastAPI, Streamlit, Groq, ChromaDB, HuggingFace

Build GPT-2 From Scratch

2025

- Built GPT-2 transformer model from scratch in PyTorch.
- Demonstrating understanding of attention mechanisms, tokenization, and autoregressive text generation (inspired by Sebastian Raschka's implementation).

Tech Stack: PyTorch, Pretrainig using Causal Masking.

Stock Price Prediction

2025

- Implemented a Stock Price Prediction system by combining Time Series Forecasting techniques with LSTM (Long Short-Term Memory) models.
- It fetches real time data from Yahoo fnance library of python.

Tech Stack: Google Colab(IDE), ML libraries(Yahoo fnance, Tensorfow, Numpy, Pandas, Matplotlib)

EXPERIENCE

Research Intern | LSTM-Based Classifier Project

Jan 2025 – May 2025

- Developed a machine learning pipeline to classify video-derived data points into predefined clusters, enabling 80% accurate and efficient data classification.
- Designed a preprocessing pipeline using one-hot encoding and sliding window segmentation to convert raw video data into time-series input.
- On going research under the guidance of Dr. Lopamudra Giri, Associate Professor, IITH.

Data Scientist | Vipani.ai

June 2025 – Present

- Built a dashboard to track the Catalogue Team's daily performance using delta and batch database processes.
- Currently developing models to predict user web journeys and purchasing behavior using Transformers.